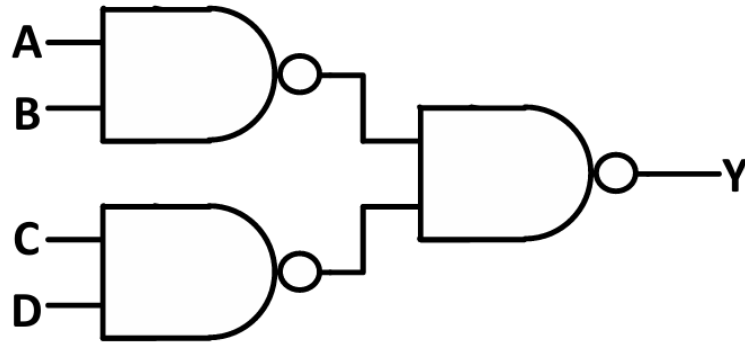


Implementation of the Boolean Function $Y = AB + CD$ Using Raspberry Pi Pico 2 W with 7447 IC



1 Problem Analysis

The given logic circuit consists of three NAND gates.

The output of the first NAND gate is

$$X_1 = (AB)'$$

The output of the second NAND gate is

$$X_2 = (CD)'$$

The final output is

$$Y = (X_1 X_2)'$$

Substituting the values of X_1 and X_2 ,

$$Y = ((AB)'(CD)')'$$

Applying De Morgan's theorem,

$$Y = ((AB)')' + ((CD)')'$$

Therefore,

$$\boxed{Y = AB + CD}$$

The output becomes logic '1' whenever:

- $A = 1$ and $B = 1$, or
- $C = 1$ and $D = 1$.

Otherwise, the output remains logic '0'.

2 Hardware Requirements

S.No.	Component
1	Raspberry Pi Pico 2 W Development Board
2	IC 7447 BCD-to-7-Segment Decoder
3	Common Anode Seven-Segment Display
4	Breadboard
5	USB Type-C Cable
6	Connecting Wires
7	220 Ω Resistors (Optional for Segment Protection)
8	Personal Computer or Laptop

3 Pin Connections

3.1 Raspberry Pi Pico 2W to 7447 IC Connections

Raspberry Pi Pico 2W Pin	7447 Input Pin	Signal Name
GP8	A	Input A (LSB)
GP9	B	Input B
GP10	C	Input C
GP11	D	Input D (MSB)
GND	GND	Ground
5V	VCC	Power Supply

Table 1: Connections between Raspberry Pi Pico 2W and 7447 IC

3.2 7447 IC to 7-Segment Display Connections

4 Hardware Setup and Accessing Raspberry Pi Pico 2 W Using Arduino IDE

The Raspberry Pi Pico 2 W is programmed using the Arduino IDE by installing the necessary board support packages and configuring the development environment. The detailed procedure is as follows.

7447 Output Pin	7-Segment Pin	Segment
a	a	Segment A
b	b	Segment B
c	c	Segment C
d	d	Segment D
e	e	Segment E
f	f	Segment F
g	g	Segment G
VCC	Common Anode	Power Supply
GND	GND (via IC)	Ground Reference

Table 2: Connections between 7447 IC and 7-Segment Display

4.1 Installing Arduino IDE

1. Download and install the latest version of Arduino IDE from the official Arduino website.
2. Open the Arduino IDE after installation.

4.2 Installing Raspberry Pi Pico 2 W Board Package

1. Open Arduino IDE.
2. Navigate to
File → Preferences
3. In the “Additional Boards Manager URLs” field, add the following URL:

`https://github.com/earlephilhower/arduino-pico/releases/download/global/package\_raspberry\_pi\_pico\_2\_w\_index.json`

4. Click “OK”.
5. Navigate to
Tools → Board → Boards Manager
6. Search for “Raspberry Pi Pico/RP2040”.
7. Install the package developed by Earle F. Philhower.

4.3 Connecting the Raspberry Pi Pico 2 W

1. Connect the Raspberry Pi Pico 2 W to the computer using a USB Type-C cable.
2. If the board is not detected automatically:
 - (a) Press and hold the **BOOTSEL** button.
 - (b) Connect the USB cable while holding the button.
 - (c) Release the button after the board appears as a removable drive.

4.4 Selecting the Board and Port

1. Navigate to

Tools → Board

2. Select

Raspberry Pi Pico 2 W

3. Navigate to

Tools → Port

4. Select the corresponding COM port of the Raspberry Pi Pico 2 W.

4.5 Uploading the Program

1. Write or paste the Arduino program into the Arduino IDE.
2. Click the **Verify** button to compile the code.
3. Click the **Upload** button.
4. Wait until the message

Done Uploading
appears in the output window.

After successful uploading of the program, the Raspberry Pi Pico 2 W executes the Boolean expression

$$Y = AB + CD$$

for all possible input combinations and displays the output through the LED connected to GP1.

5 Results Obtained in the Experiment

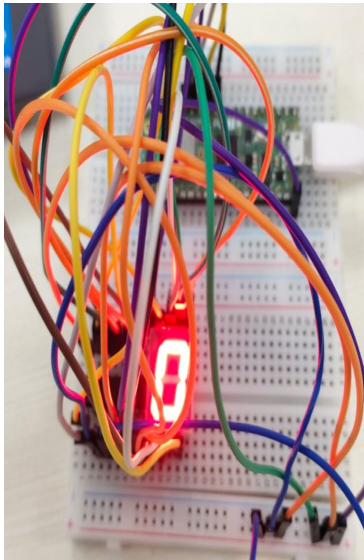
The Boolean function implemented is:

$$Y = (A \cdot B) + (C \cdot D)$$

The output was observed for all possible input combinations. The results are categorized based on the output value.

5.1 Combinations for Output $Y = 0$

The below combinations provides the output as zero.
The observations are tabulated below.



5.2 Combinations for Output $Y = 1$

